

System Design Specification

for

Employee Performance Review (EPR) Dashboard and Workflow System

Version 0.2 Approve Pending

Prepared by:

Alyssa Arroyo
Momoka Aung
Chiemela Eziechile-Nwoke
Hari Ram Gurung
Michael Hareu
Hoang Le
John Lopez
Aidan McBride
Luis Morales
James San
Hayk Vardapetyan

Santa Barbara County Public Defender's Office
California State University, Los Angeles

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Document Control

Version	Date	Author	Summary
0.1	10/16/2025	Momoka Aung, Chiemela Eziechile	Initial draft.
0.2	10/20/2025	Hoang Le (Dempsey)	Updated 3.2.3 (HR/EPR pages); added KPI defini- tions and why they matter; UX notes; added flow caption.
0.3	12/01/2025	Momoka Aung	Added detailed in- formation of System Strategy and Archi- tecture part.
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1. Introduction

1.1. Purpose

This System Design Specification (SDS) defines the complete design and implementation details for the **Intelligent HR Analytics & Employee Performance Review (EPR) Management Dashboard**, developed for the **Santa Barbara County Public Defender's Office (SBC PDO)**.

This document identifies the system components, data integrations, and automation workflows that implement the functional and non-functional requirements defined in the Software Requirements Specification (SRS). It includes detailed information on system behavior, architecture, data structures, interfaces, and component interactions.

The purpose of this SDS is to provide developers, system administrators, and stakeholders with a comprehensive understanding of how the system will be built, tested, and maintained. It is intended to serve as a technical reference throughout the development lifecycle and during future maintenance phases.

Scope of Coverage:

- Covers the entire Intelligent HR Analytics & EPR Management Dashboard, including both workflow automation and analytics components.
- Includes integration details for Smartsheet, Box.com, and Power BI.
- Excludes unrelated HR systems or legacy document storage methods not involved in this integration.

1.2. Document Conventions

This document follows a structured hierarchical numbering convention where each major section is identified numerically (e.g., 1.0, 2.0, 3.0), and subsections are represented as 1.1, 1.2, 1.3, etc.

Typographical conventions used in this document include:

- **Bold text** – denotes section titles, system names, or key terms.
- *Italics* – used for document references or emphasis.
- Monospaced text (e.g., `VariableName`) – represents field names, API endpoints, or code identifiers.

All diagrams and figures are labeled sequentially and referenced by their figure numbers. Requirements traceability follows the SRS reference IDs (e.g., “REQ-1.1”, “REQ-2.4”).

Where applicable, this SDS inherits terminology and definitions from the corresponding SRS document. Each detailed design component is directly mapped to one or more SRS requirements.

1.3. Intended Audience and Reading Suggestions

This SDS is written for a variety of technical and non-technical stakeholders involved in the project:

- **Developers** – to understand component designs, interface specifications, and data flow logic.
- **Project Managers** – to monitor design completeness, dependencies, and progress alignment with requirements.
- **Quality Assurance (QA) Team** – to validate system behavior against defined design expectations.
- **HR and Administrative Staff** – to gain an overview of system functionality and workflow automation.
- **System Integrators** – to configure Smartsheet, Box, and Power BI integrations.

Recommended reading sequence:

1. Begin with **Section 1: Introduction** for background and context.

2. Proceed to **Section 2: Design Considerations** to understand system assumptions and constraints.
3. Review **Section 3–5** for architecture and component interaction overviews.
4. Consult **Sections 6–8** for detailed design and user interface references.
5. Use **Sections 9–12** as reference material for database design, validation, glossary, and citations.

1.4. System Overview

The Intelligent HR Analytics & EPR Management Dashboard is a web-integrated solution that automates and streamlines the Employee Performance Review (EPR) process for the Santa Barbara County Public Defender’s Office. It combines workflow automation, digital document management, and analytical reporting into a unified system that supports both HR operations and strategic decision-making.

2. Design Considerations

This section outlines the major factors, assumptions, constraints, and guiding principles that affect the overall design and implementation of the **Intelligent HR Analytics & Employee Performance Review (EPR) Management Dashboard**. These considerations provide the foundation for architectural decisions, integration strategies, and technology selections made during the system design phase.

2.1. Assumptions and Dependencies

The design and functionality of this system are based on several operational and technical assumptions, as well as dependencies on external systems and environments. These include the following:

- **Cloud Infrastructure:** All major components (Smartsheet, Box.com, Power BI) are cloud-hosted and accessible via secure web connections, no on-premises servers are required.
- **Existing Licenses and APIs:** The SBC PDO HR department maintains valid enterprise or educational licenses for Smartsheet, Box, and Power BI, including API access.
- **User Environment:** End users (HR staff, supervisors, managers) have internet connectivity, browser access, and institutional Microsoft 365 accounts to authenticate with Power BI.
- **Integration Connectivity:** Secure token-based authentication (OAuth 2.0) is supported for all external system integrations.
- **Data Consistency:** Smartsheet serves as the single source of truth for all EPR tracking data, downstream systems such as Power BI and Box.com synchronize from it.
- **Organizational Change Control:** HR and IT departments will review and approve future modifications to forms, columns, and workflows before deployment.
- **Probable Enhancements:** Future revisions may include additional automation triggers or dashboards without requiring architectural redesign.

2.2. General Constraints

The following global limitations and environmental factors influence the system's design and implementation approach:

- **Platform Dependencies:** The solution relies on SaaS platforms (Smartsheet, Box, Power BI); therefore, system uptime and API availability are constrained by vendor SLAs.
- **End-User Environment:** The HR team primarily operates on Windows-based workstations using Chrome or Edge browsers; the solution must remain browser-agnostic within modern standards.
- **Standards Compliance:** All integrations must comply with SBC PDO IT security standards, HIPAA-adjacent data-handling policies, and county data-retention requirements.
- **Security and Access:** Sensitive EPR content (performance reviews, personnel records) must reside exclusively in Box.com, with folder-level access control and audit logging.
- **Performance Requirements:** Automation latency (Smartsheet triggers → Box) should not exceed typical API round-trip times within reasonable time.
- **Interoperability:** Data exchanges between systems use structured JSON payloads or CSV datasets, schema changes require configuration updates but not source-code modifications.
- **Scalability:** Designed to support the current HR staff (150–200 employees) and future growth up to 2× capacity without re-architecture.
- **Network Connectivity:** Continuous internet access is required for workflow automation and Power BI refreshes; offline operation is not supported.
- **Verification and Validation:** Testing is limited to environments provided by Smartsheet and Power BI sandboxes, no dedicated staging server exists.

- **Resource Availability:** The project team consists primarily of part-time student developers, implementation time and iteration cycles are bounded by academic schedules.

2.3. Goals and Guidelines

The following design principles and goals drive the structure, appearance, and behavior of the system:

- **Simplicity (KISS Principle):** Keep all automations, forms, and dashboards simple to understand, configure, and maintain by non-technical HR staff.
- **User Experience (UX):** The workflow and dashboards must feel intuitive, minimizing manual steps and reducing cognitive load for end users.
- **Maintainability:** Favor configuration-based solutions (Smartsheet rules, Power BI models) over custom code to ease future updates by HR administrators.
- **Transparency and Auditability:** Every EPR action from creation through approval must be traceable with timestamps and user identifiers.
- **Integration Reuse:** Where possible, reuse connectors and automation patterns across components to ensure consistency and reduce maintenance overhead.
- **Delivery Timeline:** The initial system release aligns with the end of the Fall 2025 semester; all core automations and the analytics dashboard must be operational by that date.
- **Performance vs. Cost Balance:** Optimize automation efficiency while staying within free or existing license tiers to minimize additional expenses.

2.4. Development Methods

The development and design approach for this system follows a modified **Agile Incremental Model** combined with rapid prototyping and iterative testing. Key characteristics of this approach include:

- **Short Development Cycles:** Features are implemented and validated in two-week sprints, allowing continuous feedback from HR stakeholders.
- **Early Prototyping:** Wireframes and proof-of-concept automations are built first in Smartsheet and Power BI to confirm feasibility before full implementation.
- **Continuous Integration:** Updates to Smartsheet automations and Power BI datasets are deployed iteratively; regression tests verify each change.
- **Collaborative Design Reviews:** Weekly design check-ins with the HR team and faculty advisor ensure alignment with business rules and usability requirements.
- **Adaptability:** While primarily Agile, elements of Waterfall documentation (e.g., fixed deliverables and sign-offs) are retained to satisfy sponsor requirements.

This hybrid approach enables flexibility for academic iteration while maintaining the structure and traceability expected in a professional system development environment.

3. Architectural Strategies

This section describes the major design strategies used in the Intelligent HR Analytics & EPR Management Dashboard, including technology selection, system reuse, scalability, communication, and accessibility considerations.

3.1. Technology Stack Selection

- **3.1.1** The system is built using a combination of enterprise-grade platforms — Smartsheet, Box.com, and Microsoft Power BI.
- **3.1.2** These tools were selected for their proven reliability, interoperability, and ease of automation without requiring extensive custom development.
- **3.1.3** The selection ensures compliance with Santa Barbara County’s IT policies and leverages existing agency subscriptions to minimize cost and implementation time.
- **3.1.4** Power BI provides secure and interactive visualization directly connected to Smartsheet data, ensuring near-real-time insights into EPR workflows.

3.2. Reuse of Existing Components

- **3.2.1** Instead of building a new HR platform from scratch, the design reuses existing SaaS tools already approved for use by the County.
- **3.2.2** Smartsheet serves as the workflow backbone, Box handles storage, and Power BI provides analytics, reducing duplication and ensuring data consistency.
- **3.2.3** This reuse approach accelerates deployment, minimizes maintenance overhead, and ensures seamless integration with existing County infrastructure.

3.3. Automation and Workflow Intelligence

- **3.3.1** Smartsheet automation drives the EPR lifecycle — identifying due reviews, routing for approval, and notifying HR staff.
- **3.3.2** Automation reduces manual tracking and administrative delays, ensuring that reviews are completed on time.
- **3.3.3** Automated notifications (30/14/7-day reminders) and escalations to supervisors or division leads ensure accountability and timeliness.

3.4. Integration and Communication

- **3.4.1** The system securely integrates Smartsheet, Box.com, and Power BI using native connectors and APIs.
- **3.4.2** Smartsheet end out email with attached form for signature routing, and finalized EPRs are uploaded to Box.
- **3.4.3** Power BI connects directly to Smartsheet to visualize review progress, completion rates, and overdue metrics in real time.

3.5. Modularity and Scalability

- **3.5.1** Each subsystem such as workflow, document routing, storage, and analytics, operates independently, allowing updates without impacting others.
- **3.5.2** This modular architecture supports future scaling to additional departments or new HR processes (e.g., onboarding, probation tracking).
- **3.5.3** The design can accommodate increased employee records or multi-year data without major architectural changes.

3.6. Error Handling and Recovery

- **3.6.1** Automations include validation rules to prevent incorrect data submissions (e.g., missing supervisor, invalid dates).
- **3.6.2** If Box integration fails, the system logs errors in Smartsheet and notifies HR admins automatically.
- **3.6.3** Manual override processes are available for critical tasks, ensuring that no EPR is lost or delayed due to system errors.

3.7. Data Storage and Management

- **3.7.1** All HR and EPR data are securely stored in Box.com (for finalized documents) and Smartsheet (for tracking metadata).
- **3.7.2** Data in transit between systems is encrypted via HTTPS and OAuth 2.0 authentication.
- **3.7.3** Power BI accesses only approved datasets through controlled connectors, ensuring compliance with County security policies.
- **3.7.4** Metadata tagging in Box ensures fast retrieval, auditability, and proper retention per HR policy.

3.8. Accessibility Strategy

- **3.8.1** The Power BI dashboard and Smartsheet interfaces are accessible to all users, including individuals with disabilities.
- **3.8.2** The system supports ADA-compliant color contrasts, keyboard navigation, and readable data tables.
- **3.8.3** HR and supervisors can export data into accessible Excel and PDF formats for reporting or distribution.

3.9. Security and Permissions

- **3.9.1** Access controls are role-based — HR admins have full privileges, while supervisors and division leads have limited or read-only access.
- **3.9.2** Sensitive EPR content is stored exclusively in Box, with permissions managed through County Single Sign-On.
- **3.9.3** Each integration respects least-privilege principles and audit logging for compliance.

3.10. Future Plans

- **3.10.1** Future versions may include predictive analytics for performance trends, automated reminders through Teams, or PowerApps-based self-service HR forms.

3.11. Rejected Alternatives

4. System Architecture

The Intelligent HR Analytics & EPR Management Dashboard architecture decomposes the system into manageable, interoperable components that together fulfill its core functions: automating Employee Performance Review (EPR) workflows, enabling digital signatures, centralizing document storage, and visualizing HR analytics. These responsibilities are distributed across modular subsystems that communicate through secure API integrations and platform connectors.

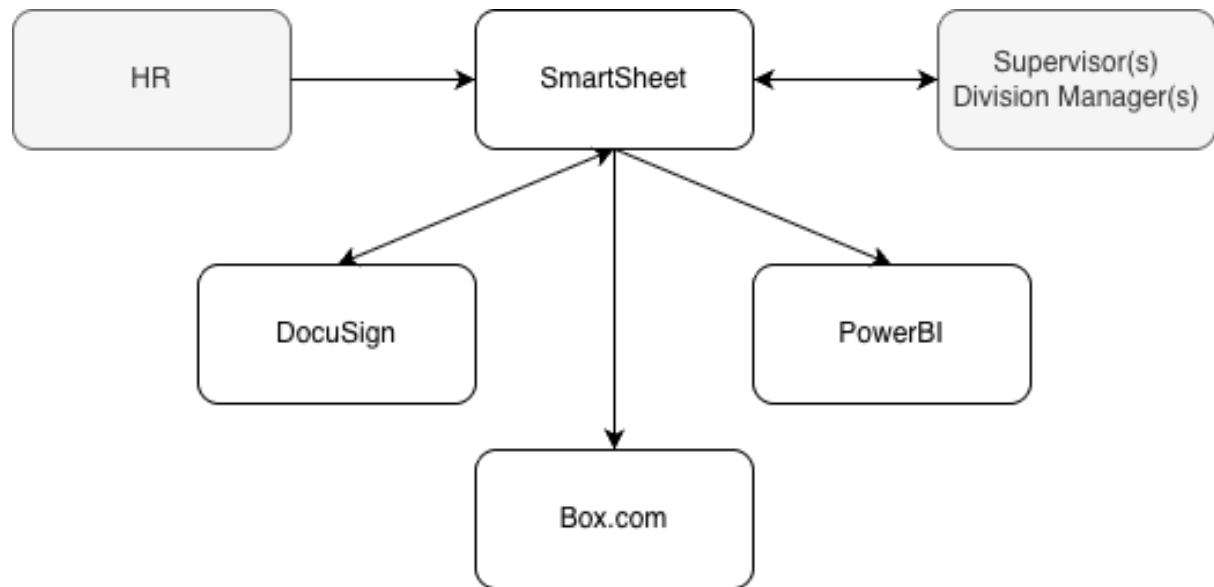


Figure 1: Project Workflow

4.1. System Decomposition

At a high level, the system is composed of several key components:

- **EPR Workflow Subsystem (Smartsheet)**

This subsystem acts as the operational hub for tracking, automating, and managing all performance review activities. It identifies upcoming reviews, assigns reviewers, tracks status updates, and triggers automated notifications or escalations. Send out email with a link to fillable form to users to get information and signatures. This ensures that review forms are securely distributed to supervisors, HR, and leadership in sequential order, maintaining audit trails and approval records.

- **Secure Document Repository (Box.com)**

Stores finalized EPR documents once they are fully approved. Files are organized automatically into structured folder paths by employee, year, and division. Access is restricted to authorized HR personnel through role-based permissions.

- **Analytics and Reporting Subsystem (Power BI)**

Provides real-time dashboards for HR staff, supervisors, and leadership. It visualizes data from Smartsheet trackers, showing EPR completion rates, overdue reviews, workforce distribution, and performance trends.

- **Integration and Data Services Layer**

This layer enables secure data flow between Smartsheet, Box.com, and Power BI. It uses native connectors, REST APIs, and scheduled refreshes to ensure synchronization and data integrity.

4.2. Component Responsibilities

Each component was designed with clear boundaries and roles to support modularity, maintainability, and security:

- **Smartsheet** manages workflow logic, employee metadata, and automation triggers for EPR creation, routing, and completion tracking.
- **Box.com** serves as the centralized document archive, ensuring all finalized EPRs are accessible and compliant with retention policies.
- **Power BI** aggregates EPR tracking data, refreshes nightly, and provides visual analytics for leadership decision-making.

4.3. Communication and Control Flow

The system follows a structured, event-driven process to manage the full EPR lifecycle.

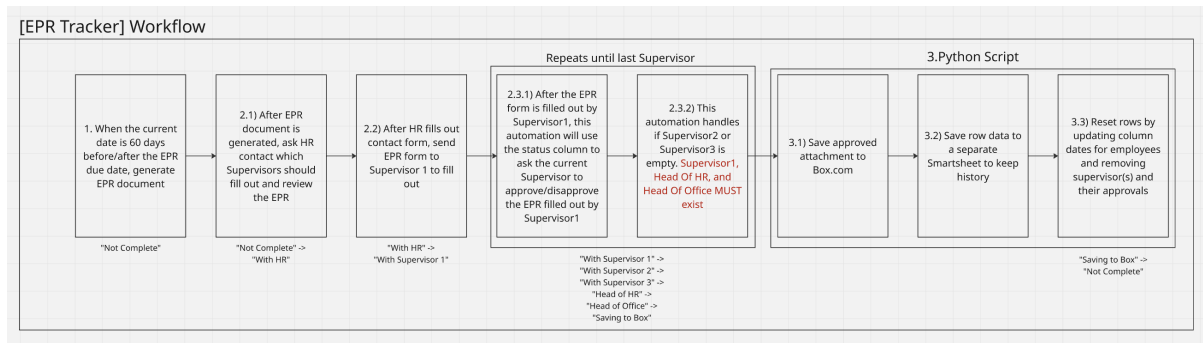


Figure 2: High-Level EPR Workflow Overview

- **1)** When an employee's review becomes due or EPR is completed, Smartsheet automation identifies the record based on the "Next EPR Due Date."
- **2)** Upon supervisor submission, Smartsheet routes the document sequentially to HR and the Head of Office for approval.
- **3)** Once all signatures are obtained, Smartsheet automatically uploads the finalized PDF to Box.com and updates the tracker with a Box URL and completion metadata.
- **4)** Power BI refreshes nightly to reflect updated completion metrics, overdue counts, and performance summaries.
- **5)** Automated emails notify reviewers of pending actions, overdue items, or completion confirmations.

4.4. Data Flow and Integration Logic

Data moves seamlessly between connected systems in a one- or two-way pattern:

- Smartsheet → Box: One-way upload of finalized documents with metadata tagging.
- Smartsheet → Power BI: One-way data synchronization for analytics and reporting.

4.5. Justification of Architecture

This modular decomposition supports several key design goals:

- **Scalability:** Each subsystem can be extended independently (e.g., additional Smartsheet trackers or Power BI pages) without disrupting the overall architecture.
- **Reusability:** The workflow and analytics patterns can be replicated for other HR functions like onboarding or training evaluations.
- **Security:** Each integration uses OAuth-based authentication and enforces role-based access in line with County IT standards.
- **Maintainability:** Low-code connectors and visual workflows reduce long-term technical maintenance and dependency on developers.
- **Transparency:** Real-time dashboards and automated logging ensure traceability across the entire EPR process.

4.6. Rejected Alternatives

- Manual PDF routing via email was dismissed because it could not provide centralized tracking or analytics capabilities.

5. Policies and Tactics

Developing the Intelligent HR Analytics & EPR Management Dashboard required a series of practical decisions and day-to-day development strategies. This section outlines the tools, practices, and policies used to ensure the system was reliable, maintainable, and aligned with county requirements.

5.1. Specific Products Used

5.1.1. Smartsheet

Used as the primary workflow platform for tracking EPR status, managing assignments, and triggering automated reminders. Chosen because it is already approved for county use and supports rapid low-code development.

5.1.2. Box.com

Used to store finalized documents with folder-level permissions and retention controls. Smartsheet holds metadata, while Box holds the actual signed PDF artifacts.

5.1.3. Power BI

Used to create HR dashboards and performance analytics. Power BI connects directly to Smartsheet and provides visual reports for supervisors and HR leaders.

5.1.4. Connectors and Authentication

Native Smartsheet, Box, and Power BI connectors were used. All integrations rely on OAuth 2.0 with least-privilege permissions.

5.2. Requirements Traceability

Requirements were tracked using a living Requirements Traceability Matrix (RTM). Each SRS requirement is linked to specific Smartsheet columns, automation rules, Box folder structures, and Power BI report pages. The RTM was reviewed each sprint, and any gaps were added to the project backlog.

5.3. Plans for Testing the Software

5.3.1. Unit Testing

Individual automations and Power BI measures were tested using mock data to ensure each component behaved correctly before connecting to live HR sheets.

5.3.2. Integration Testing

Full end-to-end tests were conducted across Smartsheet → Smartsheet → Box → Power BI. Typical scenarios included “on-time review,” “overdue reminder,” and “signature returned.”

5.3.3. User Acceptance Testing

HR staff and supervisors tested the system within a sandbox environment. Feedback was collected using structured forms and incorporated into the workflow.

5.3.4. Accessibility Testing

Dashboards were validated for color contrast, screen-reader compatibility, keyboard navigation, and readable PDF exports.

5.3.5. Data Validation

Daily checks compare Smartsheet row counts with Power BI aggregates. Any mismatch greater than 1% triggers a review.

5.4. Engineering Trade-offs

5.4.1. Low-Code vs. Custom Code

Low-code tools enabled faster delivery but limited UI customization. Custom services were deferred to avoid additional infrastructure and security reviews.

5.4.2. Real-Time vs. Scheduled Refresh

A nightly Power BI refresh was selected to minimize API usage and licensing costs. Manual refresh is available only when needed.

5.4.3. Single Source of Truth

Smartsheet is the source of truth for EPR status and task tracking. Box holds immutable signed documents. Power BI mirrors Smartsheet data for reporting.

5.5. Guidelines and Conventions

- **Naming:** Use clear, descriptive names in the format `System_Function_Description`
- **Solutions grouped logically:** Workflow, Signing, Storage, Analytics.
- **Status values normalized:** NotStarted, InProgress, PendingSignatures, Complete, Overdue.
- **Date handling:** All dates stored in ISO 8601 (UTC); localization applied only within Power BI for readability.
- **Box metadata tags:** EmployeeID, EPRPeriod, DocumentType, SignedStatus, UploadedBy, Version.
- **Change control:** All modifications peer-reviewed and logged in the traceability matrix.

5.6. Protocols

All communication across systems uses HTTPS with OAuth 2.0 authentication. Where possible, Smartsheet use webhook-style callbacks. Audit logs follow county retention policies.

5.7. Design Patterns and Programming Idioms

- **Event-driven workflow:** Due dates trigger reminders and routing actions.
- **Idempotency:** Automations check for existing envelopes or files before creating new ones.
- **Retry with backoff:** Transient API errors retry automatically up to three times.
- **Metadata-first design:** Identifier fields flow through each component to support traceability.
- **Separation of concerns:** Smartsheet for workflow, Box for storage, Power BI for analytics.

5.8. Maintaining the Software

Smartsheet sheets are backed up before major changes. Box retention policies protect archived EPR documents. Power BI datasets are monitored for refresh failures, and automation rules are reviewed quarterly with HR.

5.9. Interfaces

End users interact primarily with Smartsheet (task tracker views) and Power BI dashboards. HR administrators manage Box permissions as needed.

5.10. Source Organization

- **Smartsheet:** Employee table, EPR Tracker, and History Logs.
- **Power BI:** Data model organized by dashboard pages such as *HR Overview* and *EPR Performance*, grouped by purpose for clarity.
- **Box:** Folder hierarchy organized by Employee ID and Review Period, with controlled access permissions.

5.11. Build and Deployment Process

The system was initially tested with mock HR data. Once validated, the Smartsheet sheets, Box folders, and Power BI reports were linked to production data. Power BI reports were published to the HR workspace with a nightly scheduled refresh.

5.12. Abstraction

External systems are accessed using configurable connectors rather than custom code. Power BI measures abstract business logic so it can be reused across reports. These approaches simplify maintenance and ensure compliance with SBC PDO IT policies.

Together, these policies and tactics ensure the system remains maintainable, auditable, and secure throughout its lifecycle.

6. Detailed System Design

Most components described in the System Architecture section require a more detailed discussion. Each subsection in this section provides a detailed description of a software component within the Intelligent HR Analytics & EPR Management Dashboard. The discussion covers responsibilities, constraints, composition, interactions, resources, and interfaces for each major subsystem.

6.1. Workflow Automation Subsystem (Smartsheet)

6.1.1. Responsibilities

- Manages the creation, tracking, and routing of Employee Performance Reviews (EPRs).
- Automates notifications, reminders, and escalations to ensure timely completion.
- Generates and routes EPR documents automatically based on workflow triggers.
- Synchronizes review completion data to Box and Power BI for centralized reporting.

6.1.2. Constraints

- Dependent on Smartsheet Enterprise licensing and API availability.
- Requires stable internet connectivity for triggers, notifications, and updates.
- Automation complexity constrained by Smartsheet's conditional rule limitations.
- OAuth tokens for integrations (e.g., Box) must be periodically refreshed.

6.1.3. Composition

- **EPR Tracker Sheet** – Core tracking sheet for all employee performance review records.
- **Employee Metadata Table** – Stores supervisor hierarchy and HR data.
- **Automated Workflows and Conditional Rules:**
 - EPR Tracker 1 – Generates EPR documents at cycle start.
 - EPR Tracker 2.1 – Sends EPRs to Supervisor 1 for completion.
 - EPR Tracker 2.2 – Handles missed supervisor actions and reminders.
 - EPR Tracker 2.3 – Escalates to upper supervisors (Head of HR, Head of Office).
 - EPR Tracker 3.1 – Updates and tracks status changes during reviews.
 - EPR Tracker 4.0 – Copies finalized reviews to archival sheets.
 - EPR Tracker 5.1 – Sends periodic reminders to pending recipients.
 - TODD (EPR Tracker) – Resets and prepares sheets for the next review cycle.

Automation Workflow Diagrams The following figures illustrate the nine Smartsheet automation workflows used in the EPR tracking system. Each workflow diagram corresponds to one of the rules listed above.

Enforce probation value This rule standardizes data by setting the probationary quarter to “N/A” whenever the Employment Status is changed to “yearly.”

[EPR Tracker] 0. Enforce probationary quarter value for yearly employment type

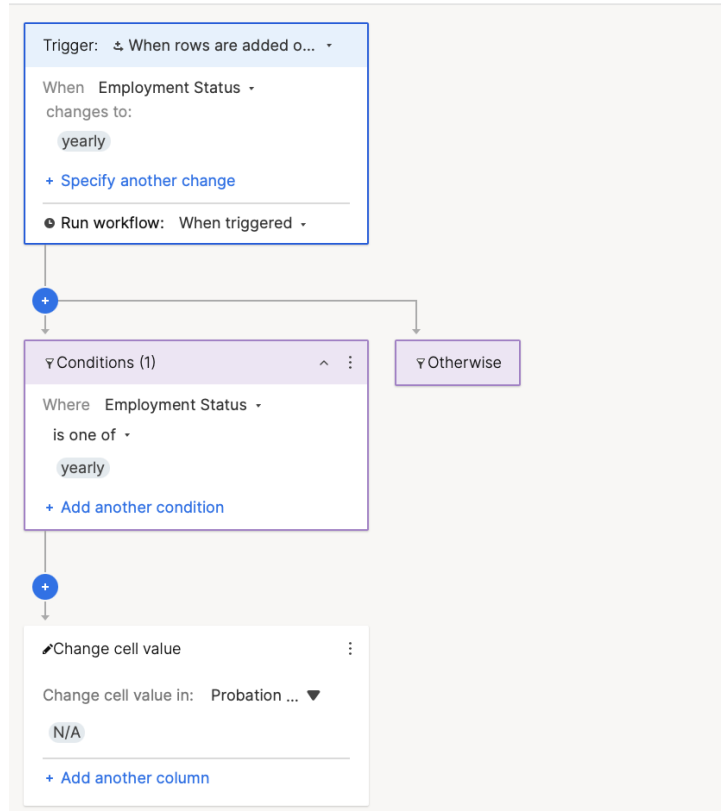


Figure 3: [EPR Tracker] 0 – Enforce probationary quarter value for yearly employment type.

Mark as probationary This rule automatically marks an employee as probationary when their employment status is set to a probationary type.

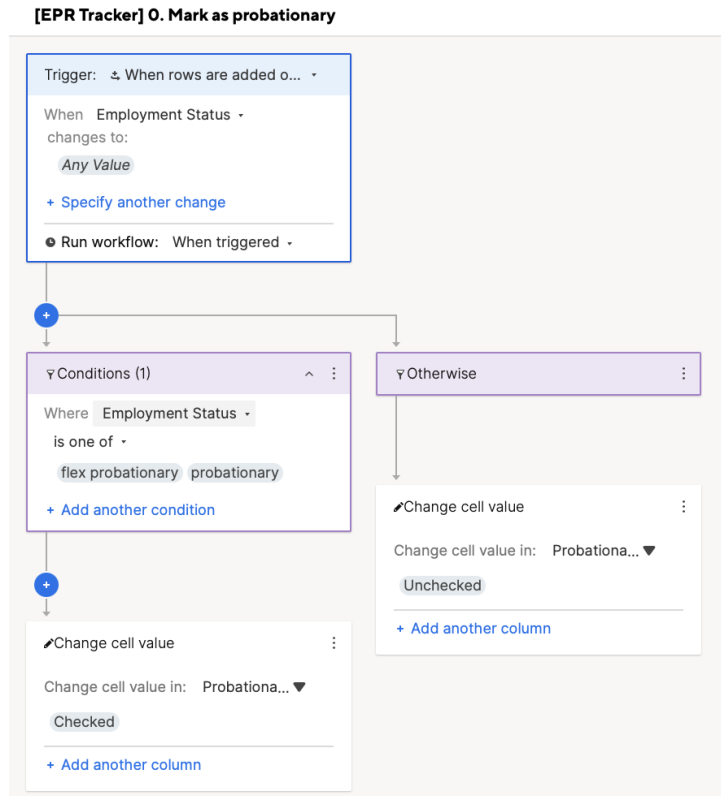


Figure 4: [EPR Tracker] 0 – Mark as probationary.

Mark EPR as late This rule flags an EPR as late when the signed due date is in the past; otherwise, it clears the late indicator.

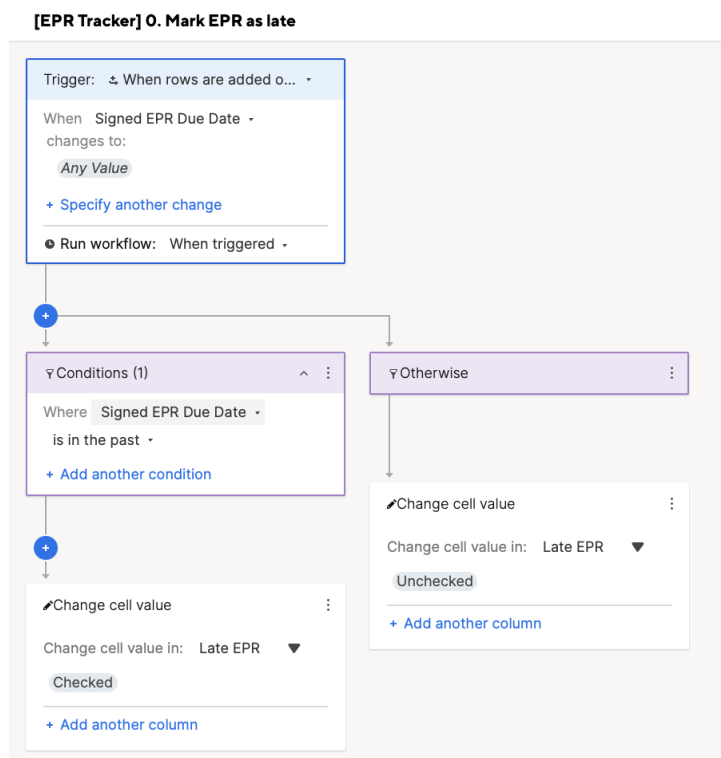


Figure 5: [EPR Tracker] 0 – Mark EPR as late.

Send reminder to holding recipients This daily rule sends reminder notifications to supervisors and HR when the signed EPR due date is approaching, based on time-window conditions.

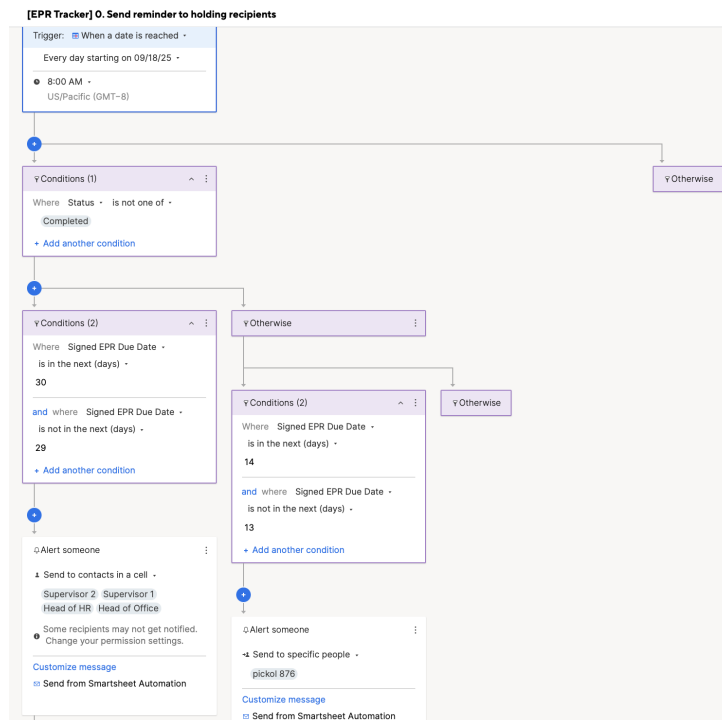


Figure 6: [EPR Tracker] 0 – Send reminder to holding recipients.

Generate EPR documents This rule creates an EPR document for an employee when their due date falls within the defined window and their status indicates the document has not yet been generated.

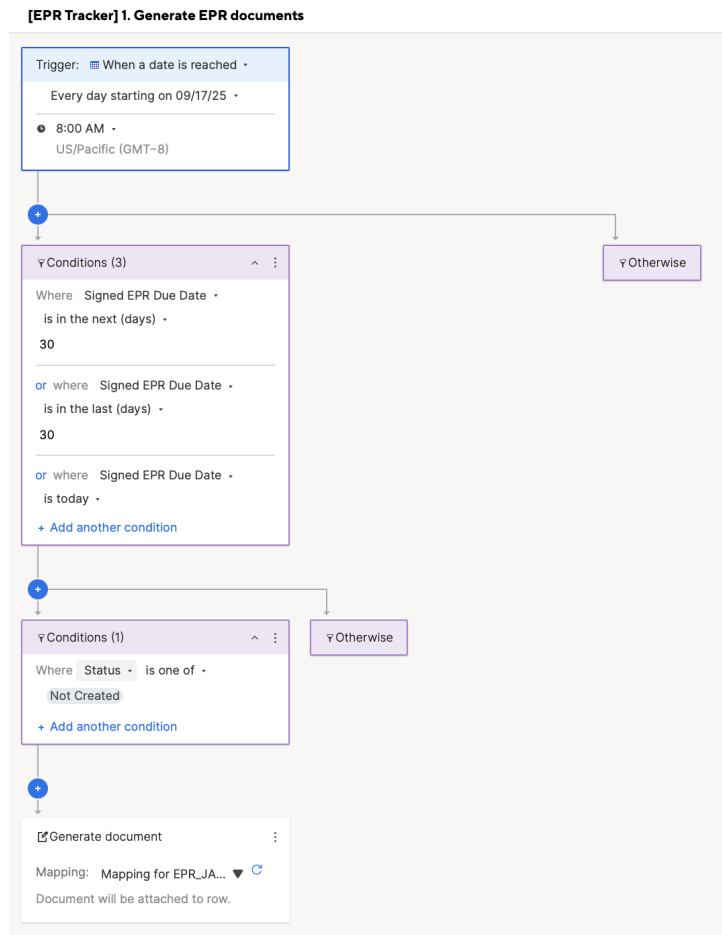


Figure 7: [EPR Tracker] 1 – Generate EPR documents.

Ask HR to fill out supervisors This rule notifies HR to enter supervisor assignments whenever an EPR document is uploaded and the review has not yet been set up.

[EPR Tracker] 2.1 Ask HR to fill out Supervisors for EPR Review

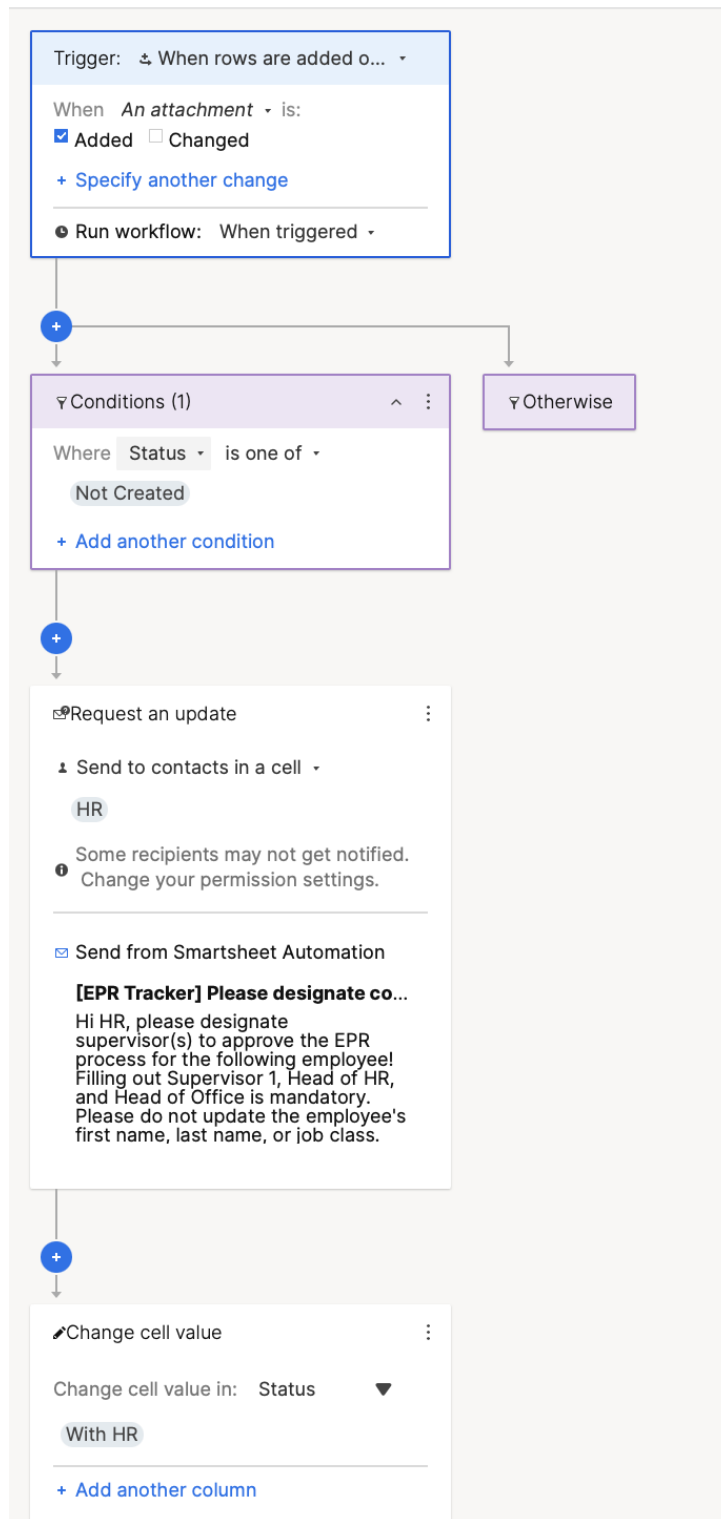


Figure 8: [EPR Tracker] 2.1 – Ask HR to fill out Supervisors for EPR Review.

Send EPR to Supervisor 1 This rule prompts Supervisor 1 to complete the EPR once HR has assigned supervisors and the review status is set to “With HR.”

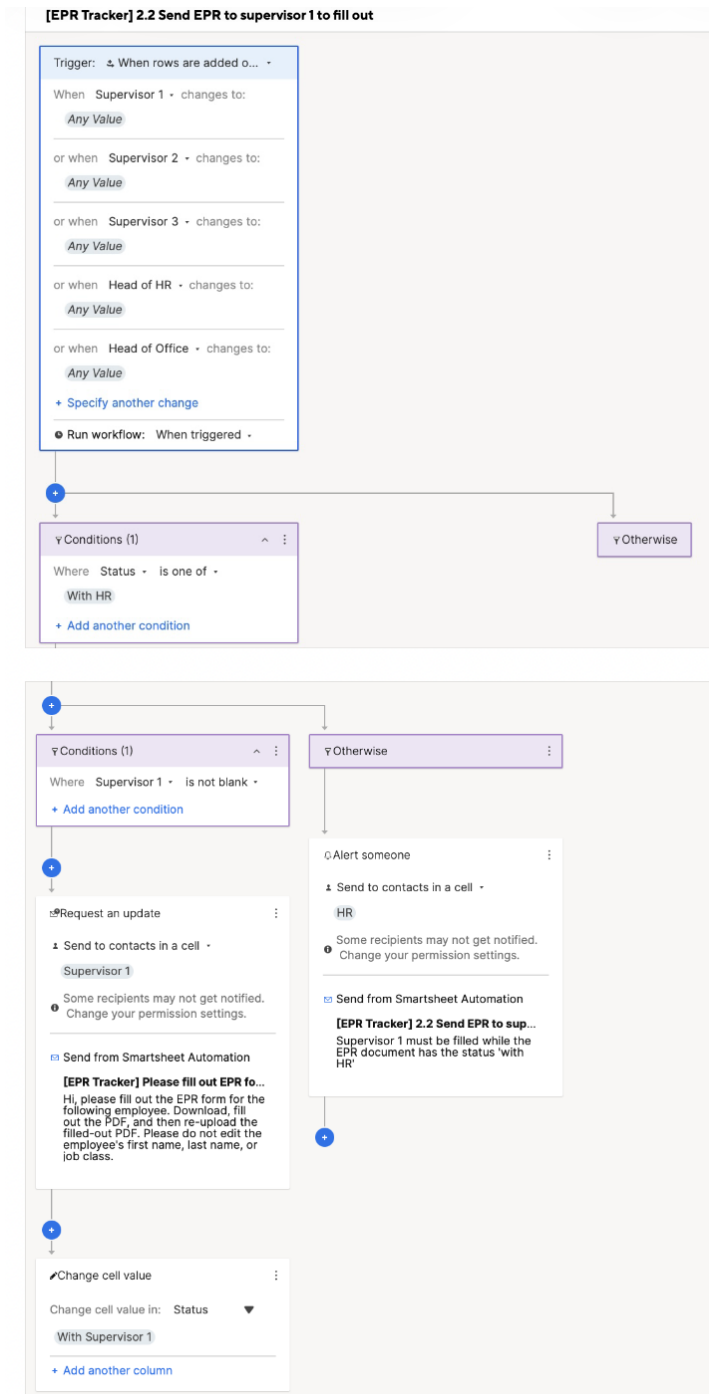


Figure 9: [EPR Tracker] 2.2 – Send EPR to Supervisor 1 to fill out.

Send EPR to upper supervisors for review This rule escalates the EPR to Supervisor 2, Supervisor 3, the Head of HR, and the Head of Office based on review status and completion conditions, ensuring the form moves through all required approval levels.

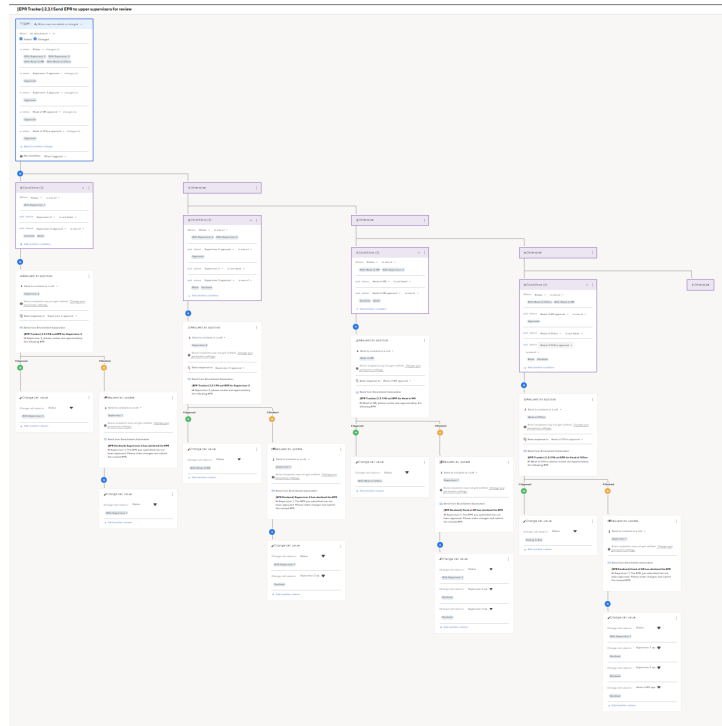


Figure 10: [EPR Tracker] 2.3 – Send EPR to upper supervisors for review.

Handle statuses when missing supervisors This rule corrects status values when required supervisor fields are missing by reassigning the EPR to the appropriate level or notifying HR if escalation targets are incomplete.

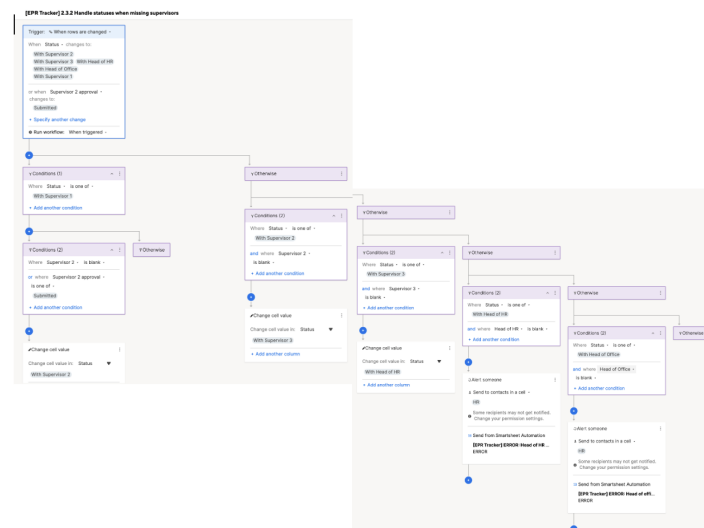


Figure 11: [EPR Tracker] 2.3.2 – Handle statuses when missing supervisors.

6.1.4. Uses/Interactions

- Sends completion updates and documents to Box for archiving.
- Pushes performance review metrics to Power BI datasets for analytics.
- Sends automated alerts to Supervisors, HR, and Heads of Office based on status changes.

6.1.5. Resources

- Smartsheet automation platform
- OAuth 2.0 authentication tokens
- Box and Power BI integration endpoints

6.1.6. Interface/Exports

- **Input:** employee records, review due dates, supervisor hierarchy, EPR attachments.
- **Output:** workflow status updates, approval timestamps, document URLs, and completion reports.

6.2. Secure Document Repository (Box.com)

6.2.1. Responsibilities

- Stores finalized and signed EPR documents in a secure, access-controlled environment.
- Maintains folder hierarchy for organized storage and retrieval.

6.2.2. Constraints

- Requires consistent metadata tagging for file placement.
- Accessible only through County-approved accounts with assigned roles.

6.2.3. Composition

- Folder hierarchy by Division → Year → Employee.
- Metadata fields for indexing and search.

6.2.4. Uses/Interactions

- Receives uploads from Smartsheet after signature completion.
- Provides file URLs back to Smartsheet and Power BI.

6.2.5. Resources

- Box Content Management API
- County Single Sign-On (SSO)

6.2.6. Interface/Exports

- Input: signed document, metadata (EmployeeID, Division, Year).
- Output: file link, storage path, upload confirmation.

6.3. Analytics and Reporting Subsystem (Power BI)

6.3.1. Responsibilities

- Provide analytics and visualization for staffing and Employee Performance Review (EPR) data.
- Deliver interactive dashboards that summarize workforce composition and EPR progress.
- Refresh data nightly from Smartsheet to ensure near-real-time accuracy.

6.3.2. Constraints

- Requires Power BI Pro or institutional license.
- Relies on Smartsheet data integrity and scheduled API refreshes.
- Access limited to authorized HR and management accounts.

6.3.3. Composition

- Page 1 – HR Overview
- Page 2 – EPR Performance Review (currently being refined)

6.3.4. Uses/Interactions

- Connects directly to Smartsheet datasets.
- Incorporates Box URLs for document verification.
- Provides visuals embedded or shared through Power BI Service.

6.3.5. Resources

- Power BI Desktop and Service.
- Smartsheet API connectors.

6.3.6. Interface/Exports

- **Input:** Smartsheet EPR Tracker data (Status, Due Date, Employee ID, Division).
- **Output:** Interactive reports, KPI cards, and downloadable summary tables.

Page 1 – HR Overview Purpose: Present the current workforce structure, job classifications, and reporting relationships.

Key Features:

- **Summary Cards (Employees, Supervisors, Interns, Employees on Probation):** Four headline tiles that update dynamically through the Find Employee slicer. *Use:* Quick reference on team size, intern cohort, and probation count for capacity checks.
- **Find Employee (Search Slicer):** Type a name to focus all visuals on that employee or group; clear the selection to reset. *Why it matters:* Enables rapid lookup without technical skill.
- **Employee Directory (Table – Full Name, Job Class, Level, Supervisor):** Read-only roster for the current selection; clicking a row filters other visuals. *Decision support:* Verify included employees and view key fields such as probation status.
- **Employees by Job Class Level:** Bar chart showing counts for Intern, I, II, III, Senior.
- **Employees by Job Class:** Horizontal bar chart showing counts by job category.
- **Combined View – Employees Count by Job Class and Level:** Ranked stacked bars combining job class and level. *Why it matters:* Helps HR visualize department composition and supervisor-to-employee ratios.

Page 2 – EPR Performance Review (Still Being Refined) Purpose: Track the status and completion of EPRs and reveal progress, delays, and workload distribution.

Key Features:

- **EPR Due Date (Between Slicer):** Date-range filter that drives all visuals. *Decision support:* Examine workload and progress for any chosen time window.
- **Job Class (List Slicer):** Filter the page to one or more job classes. *Decision support:* Compare progress across job classes or focus on a single group for follow-up.
- **KPI Cards:** Display totals for Incomplete EPRs, Completed EPRs, Total EPRs, Average Days Past Due, and Completion Rate (%). *Decision support:* Track throughput, identify overdue items, and monitor completion performance.

Notes and KPI Logic (from Smartsheet data):

- **Total EPRs** = Completed + Incomplete records in the current view.
- **Completed EPRs** = count where Status = “Completed”.
- **Incomplete EPRs** = count where Status ≠ “Completed” (includes Not Created, With Supervisor, With HR, With Heads). Previously called “Open EPRs”.
- **Overdue EPRs** = Due Date < Today and Status ≠ Completed.
- **Completion Rate (%)** = Completed ÷ (Completed + Incomplete).
- **Average Days Past Due (for Incomplete EPRs only):** Consider EPRs with Stage Group = Incomplete and Due Date < Today. For each, compute Days Late = Today – Due Date, then take the average. If none are overdue, the value is 0. Example: 2, 5, and 11 days late ⇒ Average = 6 days.

Visualizations:

- **Incomplete vs Completed EPRs by Due Month (Clustered Columns):** Monthly bars showing incomplete and completed counts for the selected range. *Decision support:* Reveal whether completions match upcoming due volume.
- **Incomplete/Past-Due EPRs per Supervisor (Horizontal Bar):** Ranks supervisors by number of incomplete or late reviews. *Decision support:* Balance workload and target follow-ups.
- **EPR Status with Total EPRs by Stage (Matrix):** Pipeline table showing counts at each stage (Not Created; With Supervisor 1/2/3; With HR; With Head of HR; With Head of Office; Completed). *Decision support:* Identify bottlenecks and responsible owners. *Helper text:* “Each stage lists who owns the number of their current EPRs.”

Interactions & UX:

- All slicers affect every visual on the page.
- Selecting a supervisor filters the Stage Matrix and monthly trend for that owner.
- Tables support export for follow-up lists.

6.4. Data Services Layer

6.4.1. Responsibilities

- Manages secure communication and synchronization among Smartsheet, Box, and Power BI.
- Ensures consistent data flow and updates across all integrated systems.

6.4.2. Constraints

- Subject to rate limits and latency from external APIs.
- Requires OAuth 2.0 authentication for each service.

6.4.3. Composition

- API integration connectors.
- Data validation and retry mechanisms.

6.4.4. Uses/Interactions

- Interacts with all subsystems for sending and receiving updates.

6.4.5. Resources

- Smartsheet API
- Box API
- Power BI REST API

6.4.6. Interface/Exports

- Input: event triggers, API requests.
- Output: validated responses and synchronized records.

6.5. Access and Audit Subsystem

6.5.1. Responsibilities

- Tracks user activity and document access across all connected systems.
- Maintains compliance with county audit and retention policies.

6.5.2. Constraints

- Depends on Box and Smartsheet logging features.

6.5.3. Composition

- Access logs and version history.
- User activity reports.

6.5.4. Uses/Interactions

- Used by HR administrators and IT auditors for compliance review.

6.5.5. Resources

- Box activity log database.
- Smartsheet version history.

6.5.6. Interface/Exports

- Input: user actions and file access events.
- Output: timestamped audit entries and reports.

Each subsystem operates within a defined scope and interaction model. This modular structure improves scalability, maintainability, and security while ensuring compliance with SBC PDO IT and data governance standards.

7. Detailed Lower-Level Component Design

This section describes the lower-level components, automation flows, and scripts that support the higher-level subsystems presented in Section 6. Each component is discussed in terms of its classification, processing narrative (PSPEC), interface description, and internal processing details, including any design hierarchy or constraints.

7.1. Processing Detail

The following diagram shows the detailed low-level logic of the EPR automation, including Smartsheet triggers, supervisor approval routing, Python post-processing, and row reset operations.

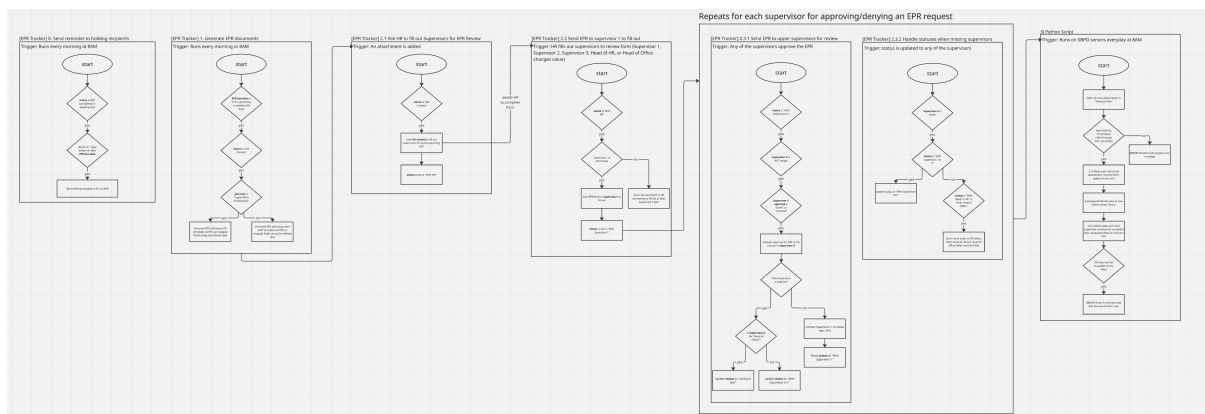


Figure 12: Detailed EPR Automation Logic (Smartsheet, Python)

7.2. Smartsheet Automation Rules

7.2.1. Classification

Configuration-based automation logic in Smartsheet that governs the routing, notification, and updating processes for Employee Performance Review (EPR) tracking. Each rule executes independently based on triggers or conditions defined within the EPR Tracker sheet.

7.2.2. Processing Narrative (PSPEC)

Automation rules are triggered when specific events occur, such as:

- Record creation at the start of an EPR cycle.
- Status changes (e.g., “With Supervisor 1”, “Approved”, “Completed”).
- Updates to assigned supervisors or approvals.
- Attachments or document uploads indicating review progress.

Upon activation, the workflows perform actions such as:

- Generating and distributing EPR forms to assigned supervisors.
- Sending reminders and escalation emails to overdue reviewers.
- Updating row statuses and progress columns.
- Triggering Smartsheet to send out email(s) to the users for signature collection.
- Copying finalized EPR data to archival or reporting sheets.

7.2.3. Interface Description

- **Input Fields:** Employee ID, Review Due Date, Status, Assigned Supervisor(s), Approval State, EPR Document Link.
- **Output Actions:** Automated email notifications, status updates, email trigger events, cell updates, and Box/Power BI synchronization.

7.2.4. Processing Detail

- **Design Hierarchy:** Flat rule-based automation structure composed of linked workflows:
 - EPR Tracker 1 – Generate EPR documents.
 - EPR Tracker 2.x – Send forms to supervisors and handle missed approvals.
 - EPR Tracker 3.x – Manage review status and escalation paths.
 - EPR Tracker 4.x – Archive or copy finalized records.
 - EPR Tracker 5.x – Schedule and send periodic reminders.
 - TODD Reset Workflow – Reset fields for next EPR cycle.
- **Restrictions/Limitations:** Executes solely within the Smartsheet automation engine; cannot perform multi-step API transactions or long-running iterative loops.
- **Performance:** Typical trigger latency under one minute, depending on Smartsheet's workflow queue and system load.

7.3. DocuSign Workflow Templates

7.3.1. Classification

Predefined DocuSign workflow templates used for routing and digital signing of Employee Performance Review (EPR) forms. Templates are triggered automatically by Smartsheet automations to initiate the review approval process.

7.3.2. Processing Narrative (PSPEC)

When triggered from Smartsheet, DocuSign generates an envelope using the corresponding EPR form template. The system automatically populates employee metadata (name, division, and review period) and routes the document sequentially through the defined approval chain:

- Supervisor 1 → Supervisor 2 → Head of HR → Head of Office

Throughout the signing process, DocuSign records timestamps, signer actions, and document versions. Once all approvals are complete, the finalized and signed PDF is returned to Smartsheet with audit trail data for storage and reporting.

7.3.3. Interface Description

- **Input Fields:** Employee name, Division, Review period, Email addresses of signers.
- **Output Actions:** Completed signed PDF, timestamp, DocuSign envelope ID, and audit trail metadata.

7.3.4. Processing Detail

- **Design Hierarchy:** Template-based configuration; each EPR type corresponds to a predefined DocuSign envelope with assigned signer routing.
- **Restrictions/Limitations:** Requires active DocuSign integration and valid signer accounts. Custom routing is limited to predefined role assignments.
- **Performance:** Envelope creation occurs within seconds; overall completion time depends on signer response intervals and DocuSign queue latency.

7.4. Power BI Data Model and Measures

7.4.1. Classification

Analytical reporting layer that transforms Smartsheet Employee Performance Review (EPR) data into structured Power BI datasets for visualization and performance analysis. The model supports real-time tracking of review progress, completion trends, and HR compliance metrics.

7.4.2. Processing Narrative (PSPEC)

A scheduled refresh process extracts data from the Smartsheet EPR Tracker sheet via the Smartsheet connector. During processing, the dataset is cleaned, normalized, and aggregated into analytical tables. The model computes several key performance indicators (KPIs), including:

- EPR completion rate.
- Number of overdue reviews.
- Average days past due.
- Departmental and supervisor-level performance summaries.

The processed dataset is published to the HR Power BI workspace and visualized through interactive dashboards accessible to management.

7.4.3. Interface Description

- **Input Fields:** Smartsheet dataset containing EmployeeID, Status, DueDate, Completed-Date, Department, and Supervisor.
- **Output Actions:** Power BI visuals, KPI cards, and performance dashboards accessible to HR and leadership.

7.4.4. Processing Detail

- **Design Hierarchy:** Flat relational data model keyed on EmployeeID and Division. Includes DAX-based calculated columns and measures for KPIs and summary analytics.
- **Restrictions/Limitations:** Dependent on Smartsheet connector schema stability and API rate limits. Field name or structure changes require dataset remapping. Refresh frequency limited to Power BI service schedule.
- **Performance:** Average dataset refresh completes within 1–2 minutes for approximately 1,000 records. Rendering of dashboards is near real-time post-refresh.

These lower-level components form the operational backbone of the Intelligent HR Analytics & EPR Management Dashboard. They handle automation, data synchronization, analytics computation, and compliance logging—ensuring that the system remains maintainable, auditable, and efficient across Smartsheet, Box, and Power BI integrations.

8. Database Design

This section describes the structure, purpose, and relationships of the primary Smartsheet data repository used by the Intelligent HR Analytics & EPR Management Dashboard. A single Smartsheet table (Employees) stores employee details, job classifications, and Employee Performance Review (EPR) workflow status. This centralized design simplifies automation, reporting, and integration with Box, and Power BI.

8.1. Employees/Employees History Records(Smartsheet)

Purpose: Serves as the master data source for all HR and EPR tracking activities. It maintains employee information, review cycles, probation data, and routing details for the entire workflow.

Field Definitions:	Field Name		Data Type	Description
	Status		Choice	Workflow stage (Not Created, With HR, Completed, etc.).
	DocuSign Status		Choice	Indicates signature progress (Not Started, Progress, Completed).
	Employee First Name		Text	Employee's given name.
	Employee Last Name		Text	Employee's surname.
	Job Class		Choice	Employee's classification (e.g., Social Service Worker, HR Manager, ADMPHS Practitioner, etc.).
	Job Class Level		Choice	Position level within the class (Intern, I, II, Senior).
	Anniversary Month		Choice	Month when employment or review cycle begins.
	Probationary EPR		Checkbox	Marks whether an employee is in a probationary review cycle.
	Probation Quarter		Choice	Quarter of the probationary cycle (1Q–4Q, Annual).
	Probation Due Date		Date	Due date for probationary reviews.
	Signed EPR Due Date		Date	Target completion date for the employee's current EPR.
	Previous EPR Signed		Date	Date the last EPR was signed.
	Previous EPR Actual Due Date		Date	Original due date for the last EPR.
	Supervisor		Text	Primary supervisor responsible for the employee.
	HR		Text	Assigned HR reviewer verifying EPR progress.
	Head of HR		Text	HR leadership reviewer for final approval.
	Head of Office		Text	Department or division head providing final sign-off.
	Supervisor 1 / 2 / 3 Approvals		Checkbox	Indicates approval completion at each supervisory level.
	Head of HR / Head of Office Approvals		Checkbox	Indicates final review and approval stages.
	Reset Ready		Checkbox	Marks record as ready to restart next review cycle.
	Late EPR		Checkbox	Flag indicating whether the EPR is overdue.

Constraints:

- Each employee appears once per active review cycle.
- Approval steps must follow the set sequence (Supervisor → HR → Head of HR → Head of Office).

of Office).

- Only authorized HR users can modify workflow status, DocuSign Status, or approval fields.
- All date values must follow ISO format (MM/DD/YY) for consistency with Power BI refresh logic.

8.2. Data Relationships and Integration

- The Smartsheet table connects directly to Power BI via the Smartsheet connector for analytics and reporting.
- Box integration automatically uploads completed EPR documents and writes the storage link back into the table.
- Power Automate handles synchronization across Smartsheet, Box, and Power BI.

8.3. Security and Access Control

- Smartsheet enforces role-based permissions: HR Admins (edit all), Supervisors (edit assigned rows), Employees (view only).
- All edits are version-tracked and timestamped automatically for compliance.
- Data access through APIs is secured using OAuth 2.0 authentication.

This single-sheet data structure provides modularity, real-time synchronization, and high traceability while maintaining compliance with SBC PDO HR data governance policies.

9. User Interface

9.1. Overview of User Interface

From the user's perspective, the Intelligent HR Analytics & Employee Performance Review (EPR) Management Dashboard is a set of connected cloud interfaces that together automate and visualize the county's HR review process. Users interact with the system primarily through four existing platforms: **Smartsheet**, **Box**, and **Power BI**.

- **Smartsheet** acts as the operational workspace where HR staff and supervisors manage employee data, review schedules, and workflow progress.
- **Box** hosts finalized and signed EPR documents within structured, access-controlled folders.
- **Power BI** delivers interactive analytics dashboards for HR leadership to track review completion, overdue items, and workforce trends.

Each interface is user-friendly and designed for non-technical users. HR staff receive an email notification from Smartsheet when an employee's EPR is due in 60 days. HR then enters the employee's supervisor(s), assigned HR reviewer(s), and department head into the system. Supervisors receive Smartsheet email notifications prompting them to review and sign digital forms. Once completed, the forms are automatically routed to HR for verification. Box folders are organized by division and year, ensuring that all finalized documents are retrievable without additional setup. Power BI visualizations present summarized results through charts, KPIs, and slicers for real-time performance insights.

Accessibility and usability are key considerations. All platforms support high-contrast modes, screen-reader compatibility, and responsive layouts across desktop and tablet devices. The modular use of familiar commercial interfaces minimizes training needs while providing a consistent experience for both HR personnel and supervisors.

9.2. Screen Frameworks or Images

- **Smartsheet:** Employee Table and EPR Tracker sheets containing employee names, job classes, probationary data, due dates, and review statuses.
- **Box:** Folder directory by Division → Year → EmployeeID storing finalized and signed EPR PDFs.
- **Power BI:** HR Overview page showing workforce summaries and job class distributions; EPR Performance Review page showing completion metrics, overdue counts, and supervisor workload.

9.3. User Interface Flow Model

The user interface operates as a connected workflow rather than a single web portal:

1. **Initiation:** HR staff create or update employee entries in Smartsheet.
2. **Routing:** Automated Smartsheet triggers send an email attached EPR forms for HR staff and supervisor(s).
3. **Completion:** Once all parties sign and approved, Smartsheet routes the finalized document to Box, where it is archived in the correct folder path.
4. **Analytics:** Smartsheet data refreshes nightly in Power BI, updating KPIs, completion rates, and overdue alerts.

Each stage provides immediate visual feedback: status updates and email confirmations in Smartsheet, upload verification in Box, and live metrics in Power BI. This integrated, user-centric design streamlines the entire EPR process, enabling HR staff and supervisors to track, approve, and analyze reviews efficiently across familiar cloud interfaces.

10. Requirements Validation and Verification

This section describes how the main functions of the Intelligent HR Analytics & EPR Management Dashboard are tested and verified to meet system requirements.

Requirement	Component / Module	Method of Testing
The system provides a simple and accessible user interface for HR and supervisors.	User Interface (Smartsheet, Power BI, Box)	Manual walkthrough and user feedback sessions; accessibility and usability review.
The system allows HR staff to create, edit, and track employee performance reviews.	Smartsheet Automation Rules	Functional test by adding and updating employee rows; checking rule triggers.
The system routes EPR forms for signatures in correct order.	DocuSign Integration	Integration test with sample forms; verify signing order and completion notices.
The system saves completed EPRs automatically in Box.	Box Storage Connector	Upload verification; folder name and file access check.
The system displays EPR status and completion data in Power BI.	Power BI Dashboard	Compare dashboard counts with Smartsheet data; test slicers and visuals.
The system updates Power BI data each night.	Data Services Layer	Confirm daily scheduled refresh and timestamp in dataset.
The system enforces user roles and permissions.	Smartsheet / Box Permissions	Test edit rights for HR, Supervisor, and Employee roles.
The system protects data with encryption and secure access.	Smartsheet, Box	Verify HTTPS connection and County IT compliance.
The system handles multiple users without performance loss.	Power Automate / Smartsheet	Simulate concurrent updates; observe response times.
The system records activity and changes for auditing.	Smartsheet History Log	Review logs for date, time, and user accuracy.

All major modules were verified through user testing and workflow simulations to confirm that the dashboard operates correctly, securely, and efficiently.

11. Glossary

- **EPR (Employee Performance Review):** A structured evaluation process for employee performance and development.
- **HR (Human Resources):** The department responsible for employee records, reviews, and workflow management.
- **Smartsheet:** Cloud-based platform used to manage employee data, workflow automation, and EPR tracking.
- **Box:** Cloud storage system used to archive finalized and signed EPR documents.
- **Power BI:** Data analytics tool used to visualize EPR metrics, progress, and workforce summaries.
- **Power Automate:** Automation service used to link Smartsheet, DocuSign, Box, and Power BI for workflow synchronization.
- **Dashboard:** A Power BI visual interface displaying HR analytics and EPR tracking metrics.
- **OAuth 2.0:** Authentication method used to ensure secure access between connected services.
- **KPI (Key Performance Indicator):** Quantifiable metric used to measure progress toward HR goals.
- **SRS (Software Requirements Specification):** The document that defines all functional and non-functional system requirements.
- **SDS (Software Design Specification):** This document; describes how the system is structured, integrated, and implemented.

12. References

- Software Requirements Specification – Intelligent HR Analytics & EPR Management Dashboard
- Smartsheet Product Documentation: <https://help.smartsheet.com/>
- Microsoft Power BI Documentation: <https://learn.microsoft.com/power-bi/>
- Microsoft Power Automate Documentation: <https://learn.microsoft.com/power-automate/>
- Box API and Security Overview: <https://developer.box.com/>
- Santa Barbara County Public Defender's Office – HR Process Guidelines (Internal Reference)
- CSULA Senior Design Project Portal – Team Sun-1 Documentation (2025)